Mill Plan

Milling sequence and setup instructions

Board Specifications

Dimensions: 12" (305 mm) × 18" (457 mm) × 1-1/2" (38 mm)
Style: checkerboard
Species: Walnut, Maple

Figures: see the cut sheet PDF (this mill plan is text only).

Milling Sequence

1. Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3.2 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

- Final pass depth: ~1/64" (0.4 mm)

2. Rip strips

Rip strips with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

- Fence to blade: 1-1/16" (27 mm)

3. Glue up

Glue the first edge-grain panel. Alternate clamps above and below the panel and check for flat with a straightedge.

- Clamp spacing: about 8" (203 mm)

4. Flatten first panel

Flatten Panel A, sub-panel 1/2, 6-1/4" (158 mm) wide:
Strip-set (long grain, before slicing): Planer OK: 6-1/4" (158 mm) wide, long grain.
Flatten Panel A, sub-panel 2/2, 6-1/4" (158 mm) wide:
Strip-set (long grain, before slicing): Planer OK: 6-1/4" (158 mm) wide, long grain.

5. Crosscut slices

Crosscut the panel into slices of the final board thickness plus the end-grain flattening allowance (5 mm), on a crosscut sled with a stop block.
Slice size: 1-11/16" (43 mm).
Final thickness: 1-1/2" (38 mm).
End-grain allowance: 3/16" (5.0 mm).
Slices: 36 in all (Panel A, sub-panel 1/2 18, Panel A, sub-panel 2/2 18).
Panel A, sub-panel 1/2 (order: W M W M W M): sled must clear 6-1/4" (158 mm).
Panel A, sub-panel 2/2 (order: W M W M W M): sled must clear 6-1/4" (158 mm).

- Slice length: 1-11/16" (43 mm)

6. Rotate slices

Turn each slice end grain up.

7. Flip alternate slices

Flip every other slice end for end so the colours alternate into squares.

8. Glue up

Glue slices into the final panel, aligning the squares at every joint. Alternate clamps above and below the panel and check for flat with a straightedge.

- Clamp spacing: about 8" (203 mm)

9. Flatten and sand

Flatten the final glue-up, 12" (305 mm) wide:
end grain: router sled or drum sander, never the planer.
Then sand: 80, 120, 180 grit. Break edges (180 grit).

- Grits: 80 / 120 / 180

10. Edge profile

Rout the edge profile, default: 1/8" (3.2 mm) roundover. Optional finger grips: core box / round-nose bit 3/8-1/2" (9.5-12.7 mm).

- Bit: 1/8" (3.2 mm) roundover

11. Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.